

565 IR STARK SHIFT THINGY

THE AUTHOR

ABSTRACT. Our chosen system of study is the 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3-benzothiadiazol-4-yl)methylene]propanedinitrile (DTDCPB) chromophore prevalent in organic photovoltaic materials research. DTDCPB has nitrile groups that make it ideal for IR studies due to its strong signature stretching frequency. For the first component of our project, we hope to simulate some ground state IR spectra of DTDCPB in the presence of varying external fields. This can then be used to calculate the Stark shift and build a calibration curve for the effect of external field on the Stark shift of this molecule. Furthermore, we plan to probe the excited state of DTDCPB as well and see if we can look at Stark effects in the excited state as well. The third step we hope to investigate is a pump-probe transient IR study of this molecule where the excited state dynamics of DTDCPB can be observed. Hopefully our understanding of Stark effects in the ground and excited states will elucidate some of the dynamics of the transient IR study.

1. INTRODUCTION

As research in solar energy becomes increasingly important for environmental applications, several new strategies are being investigated to increase the efficiency of these devices. Specifically in organic photovoltaics (OPVs), one such strategy has been to use small molecules with large ground-state dipole moments. It is believed that these dipoles can allow for enhanced hole hopping and can minimize charge recombination (ref Griffith). One particularly promising molecule is 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3-benzothiadiazol-4-yl)methylene]propanedinitrile (DTDCPB) and has resulted in OPVs with power conversion efficiencies of up to 9.6%. The nitrile groups present in the molecule can act as reporter moieties for vibrational spectroscopy. By modeling the effect of an applied electric field on absorption spectrum (Stark Effect), it is then possible to gather information regarding the local electric field during charge transfer events in a photovoltaic device via excited state IR spectroscopy. The first requires a thorough understanding of the electric field dependence of the ground state IR spectrum before probing the excited state.

2. THEORY

Some theoretical hand-wavey bullshit and lots of "beautiful mathematics" shall go here. Someday.

Date: May 18, 2016.

3. METHODOLOGY

4. COMPUTATIONAL RESULTS

5. DISCUSSION

6. CONCLUSION